



Full Length Article

Neural network-based control of an ultrafast laser

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ABSTRACT

With the recent advances in machine learning (ML) and data science (DS), the control, modeling, and analysis of these complex systems continues to improve. In this work, we report on the optimization of the intensity of a femtosecond laser using feedforward neural networks (FFNN) that model the input–output relationships of the data. The input parameters of the system were optimized to achieve the required performance of the femtosecond laser. We propose a neural network-based control system to model the relationship between the spectral amplitude and phase of the input laser pulse at the amplifier input and the shape of the output pulse. Low-jitter laser parameter inputs and the resulting laser pulse duration were modeled, and the resulting correlation between the input and output data was used to optimize the laser pulse. We demonstrate improved processing and laser control performance.

1. Introduction

In recent years, machine learning (ML) has attracted immense attention in the research community [1]. Several recent reports, including the “AI for Science” from the Department of Energy [2], detail the level of interest of the scientific community and outline directions that need to be taken. The versatility of ML models and tools allows them to handle high complexity, thereby leading to better optimization and control. As described in the literature [1], ML models are relevant to almost all scientific fields. ML models have been at the forefront of many research areas including, computer engineering [3], life sciences [4], and physics [5]. Computer vision, materials science, financial engineering, and biotechnology are some of the areas where ML models have been extensively used for data classification and prediction [1,3,4,6]. The concept of Neural Networks (NNs) is the foundation of the ML techniques used today. A neural network is a well-known structure composed of multiple layers of nodes, each of which performs an affine transformation of the output of the previous layer, followed by a nonlinear activation function, which is typically a sigmoid or a rectified linear unit (see, e.g., [7]) to perform ML tasks. With the advent and availability of computing facilities, such as the Argonne Leadership Computing Facility (ALCF), the training of NNs on large amounts of data has become possible and commonplace. Here we are seeking to apply ML to research infrastructures, namely particle

accelerators [8] and specifically the laser driving a genre of particle accelerator.

Today, we tend to build particle accelerators based on radio-frequency (RF) cavities. There is a fundamental limit to how high an acceleration gradient can be with metallic normal-conducting copper walls or the superconducting (Nb) walls of the cavity before it breaks down due to the high electric fields. By using only oxygen-free copper and applying silver coatings, these gradients can be pushed higher in the normal-conducting case. For example, recent literature reports that this limit is about ~100 MV/m [9] for typical conducting cavities at room temperature, after which the copper-only, silver-coated cavity exhibits electrical breakdown. For superconductors, the typical cavity gradient at 2 K is about 30 MV/m. With such a constraint, we can only practically achieve high-energy particles in a large accelerator system. These systems, such as those in high-energy physics and free-electron laser research infrastructures, cost well over a billion US dollars [10].

Recent research illustrates that a 100-km-long accelerator is needed to produce energetic particles to study unusual phenomena in high-energy physics [11]. Now if we think about the next generation of linear colliders, we need these larger machines on the order of 100 km per LINAC (one electron and one positron). It would of course be useful to have smaller, more compact accelerators for a variety of applications, including free-electron lasers and this next linear collider [12]. Thus, a major challenge in the accelerator field is to produce high-energy

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particles with a smaller accelerator. We also hope that a more compact approach would result in a much more cost-effective and competitive method in terms of beam quality, stability, and operational availability than RF-based accelerators. Using lasers to drive particle accelerators is one attractive approach to more compact accelerator systems. Here we seek to apply machine learning (ML) to a laser for driving a particle accelerator.

Laser physics is one of the areas where ML-based approaches and research are still in their infancy. Control systems are of paramount importance, especially when laser systems are complicated and consist of multiple networks of laser systems. Therefore, there has been immense engineering interest in trying to improve the various control systems traditionally used in different laser systems [13]. However, several challenges need to be addressed, such as unpredictable environmental changes, nonlinear material dynamics, and the inherent noise in the system. Linear systems, which are easier to implement and manufacture with theoretically high performance, fail to address such challenges when implemented in practice. ML can help with fast data collection and processing and computational tasks to solve some of these challenges. For example, AI can help maintain data integrity by detecting and discarding anomalies or manipulated data [14], or to automatically categorize the data [15]. For example, semi-supervised algorithms have been developed to help in fast automatic labeling of newly acquired data [16].

A main area of laser physics where ML techniques is in great demand are the control techniques used in these complicated systems. In this regard, the use of neural networks has helped in the diagnosis and analysis of various complex, time-varying, and nonlinear systems present in ultrafast laser control and has helped in modeling the function of these machines [17,18]. Therefore, to deal with the nonlinearity and complexity of a system, ML is a suitable tool that is flexible and efficient for the optimization process.

In this study, we design a laser control model using ML techniques based on NNs. In this context, the Center for Ultrafast Optical Science (CUOS) at the University of Michigan has previously reported research efforts on the adaptive control of the laser system and LWFA (Laser Wakefield Acceleration) [19–21] by using a genetic algorithm. We implemented the NN's control systems using laser parameter inputs controlled by the DAZZLER [22] under open-loop conditions, with the corresponding laser pulse shape outputs measured by the frequency resolved optical gating (FROG) [23] to train the network. Moreover, this model has been constructed using a NN architecture, and the detailed result, along with a comparative analysis between the proposed deep controllers, has been performed to demonstrate the performance, effectiveness, and response time of the proposed technique. Here, FROG is used as a diagnostic tool to retrieve the phase and amplitude of the pulse to be retrieved simultaneously. This work presents new opportunities to further explore the applications of ML, NNs, and deep learning in the context of laser systems, including those for driving particle accelerators.

2. Motivation and prospective study

The performance of the system we want to improve by optimization depends on several key parameters, i.e. 2nd, 3rd, and 4th-order phases, which are essential for preventing of the laser-irradiated target from being destroyed. We need to understand how their variation affects the overall system performance. In addition, the cost of computation and the architecture of the NNs must also be considered in the optimization. Despite many possible and real challenges in the implementation and use of NNs such as inaccessible data and data security, infrastructure requirements for testing and experimentation, etc. [24], these techniques have already been proven to be useful in laser science. For example, ML and NNs, have been used to collect and analyze data for high-energy physics experiments that produce particles with particle accelerators [25]. These NN-driven techniques have allowed the

statistical analysis and detection of particles and electrons for further identification.

To set the stage, consider an example of where improvements in control can improve the results of the application. For example, the Linac Coherent Light Source (LCLS) [26] and the future machines LCLS II and LCLS II HE at SLAC National Accelerator Laboratory [27] are enabling and will enable new investigations of high energy density (HED) physics. For example, the Matter in Extreme Conditions (MEC) facility, which uses the LCLS X-rays as a probe, continues to explore of HED plasmas generated by long- and short-pulse lasers. This has allowed studies of the structure and properties of warm dense matter, nanometer-scale plasma features, dynamic phase transitions, lattice dislocations, and shock waves in materials [28]. Diagnostics at the MEC allow direct measurement of the material properties of each shot [29]. In these experiments, the brightness of the X-ray free-electron laser (FEL) is critical to obtaining the resolution in the experiments. A bright electron beam is essential for the generation of the X-rays by the FEL process [30] in this SRF accelerator system. More advanced control of the accelerator, FEL, and facility's laser systems could improve operations as well as experimental outcomes.

Looking toward a laser-driven accelerator system, preliminary reports suggested that there might be an enhancement in the peak charge density by a factor of 20 compared to the electron beam profile with the help of a feedback control system based on a genetic algorithm (GA) [20]. This was in line with the enhancement of charge density by factors of 20 using a GA with the initial profile optimization of the electron beam, as reported by He et al. [20].

This example of potential improvements to an overall laser system led us to explore the use of neural networks (NNs) on a laser for possible application to driving accelerator systems. NNs are already being used in LWFA as demonstrated in a recent study [31]. As mentioned above, we applied our methods to the optimized results from CUOS at the University of Michigan [20]. In line with the motivation, the experimental setup included the use of the output image from a charge-coupled device (CCD) when a 30-V signal was applied to 37 actuators. The experiment conducted by He et al. [19] required the final solution to be obtained using a GA for all 37 voltages. The details of the experiments can be found in the previous publications [19,20]. However, the GA-based approach requires a long time to iterate the input parameters to find an optimized output. Genetic algorithms calculate the fitness function repeatedly to find a good solution. Therefore, it takes a good amount of time to compute a reasonable solution. Pre-trained neural networks generally take much less time for new input. Conducting experiments for each voltage is cumbersome because changing the voltage of each actuator and repeating the measurement takes a significant amount of time. Data acquisition would be challenging because of the need for real-time processing.

The purpose of this work is to investigate the relationship between the input and output signal parameters of the available control and measurement instrumentation for the femtosecond laser for wakefield acceleration. We were not able to obtain the exact parameters from He et al. [19], so we proceeded with a different data set for a slightly different experiment. In our approach, the primary optimization of training and testing was considered using inputs that followed the open-loop conditions. The final output of the open-loop training and testing process was then used for consistent prediction to train the NN. This approach will help us to demonstrate the improved performance of their laser system, including the improved performance of this and other laser-driven particle accelerator facilities. We have also identified the methods and discussed their feasibility, along with the different numbers of datasets required and infrastructure strategies in the next section.

3. Research methodology

The research methodology of this work can be summarized with the following steps: (1) development of the NN-based models, (2) demonstration of the control using the NNs, (3) laser modeling, (4) training of the NNs and (5) testing the model.

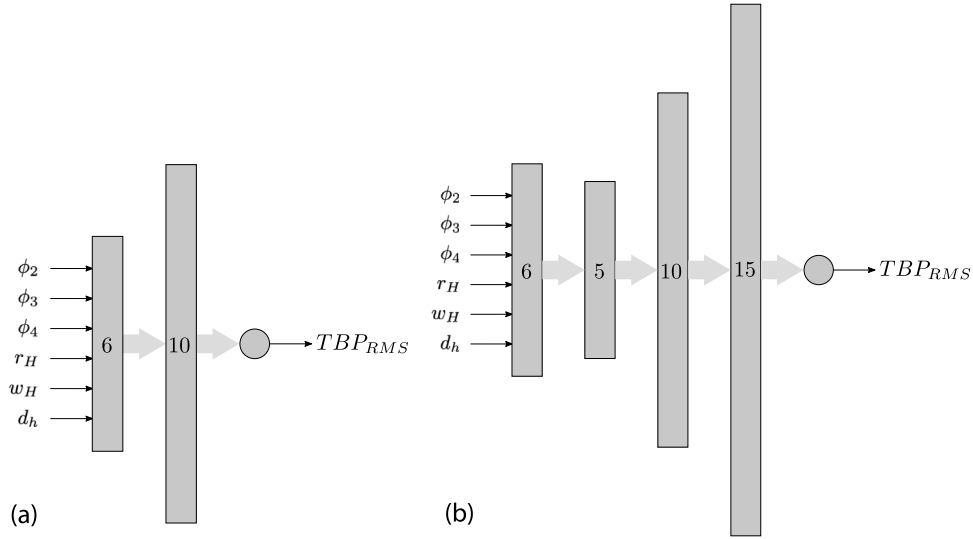


Fig. 1. Schematic representation of the neural networks. The first model (a) has one hidden layer with 10 neurons and the second one (b) has 3 hidden layers with 5, 10, and 15 neurons, respectively.

3.1. Development of the models

To solve the regression problem, we used an approach based on the neural networks implemented in the MATLAB environment [32–35]. In this study, we used two models denoted as NN1 and NN2 as schematically shown in Fig. 1. NN1 is implemented with the help of `fitrnet` function.

Both models have an input layer that takes six features, and the last layer produces the estimated output. In the intermediate hidden layers, we used the rectified linear unit (ReLU) activation function, but in the last layer, we used a hyperbolic tangent (Tanh) function [36]. Each neuron of the fully-connected layer is assigned a vector of weights that are optimized during the training process.

For the NN2 network, we chose the Levenberg–Marquardt algorithm [37] for training, which combines two numerical minimization algorithms: the gradient descent method [38] and the Gauss–Newton method [39]. The latter takes the sum of the squared errors, which is reduced by assuming that the least-squares function is locally quadratic in the parameters. It also determines the minima of this quadratic function. The gradient descent method takes the sum of the squares of the errors and reduces them using an iterative method. The parameters are initialized and updated in the direction of the steepest descent of the curve. The combination of the two methods is called the Levenberg–Marquardt method. The gradient descent method is used when the parameters are far from the optimal value. When the parameter values are closer to the optimal value, the Gauss–Newton method is used.

3.2. Demonstration of control

Our experimental setup is based on the Ti:sapphire laser system named “ λ^3 ”, which delivers 35 fs, 20 mJ laser pulses at a repetition rate of 480 Hz (see the photo in Fig. 2). The pulses are generated by two separate multipass amplifier chains pumped by commercial Nd:YLF lasers and compressed after amplification by a high-throughput transmissive holographic grating compressor. We take advantage of the high repetition rate of this laser system to perform statistical analysis of various experiments that are not readily available for most high-intensity laser systems, which typically have repetition rates on the order of 1 Hz. The primary goal is to use this high repetition rate to allow us to perform optimization routines for a wide range of laser-plasma interaction experiments using adaptive optics and ML techniques.

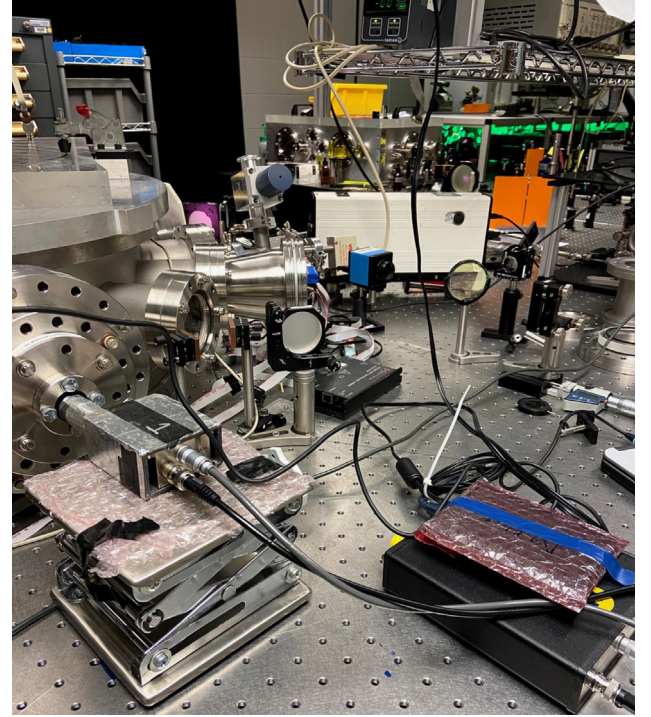


Fig. 2. Our experimental setup at CUOS at the University of Michigan.

Previous work using “ λ^3 ” involved a deformable mirror and a GA to optimize a laser wakefield acceleration experiment [20]. Considering the motivation of this study, we aimed to use a higher-order optimization algorithm for different laser parameters. The goal is to improve the laser pulse compression using a DAZZLER acousto-optic programmable dispersive filter [22] and the aforementioned neural networks. The data used for optimization were obtained from three different sources: the femtosecond laser cavity, the DAZZLER (pulse shaping), and the FROG (diagnostic), as shown in Fig. 3. The DAZZLER is an ultrafast pulse shaping system that performs simultaneous and independent spectral phase and amplitude programming of ultrafast laser pulses. The femtosecond laser cavity provides a raw laser pulse that can be modified with the DAZZLER. The six DAZZLER settings:

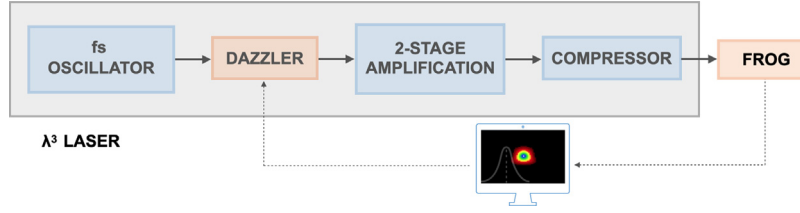


Fig. 3. Schematic of the experimental setup based on the λ^3 laser, with the DAZZLER pulse shaper and the FROG diagnostic feedback.

2nd, 3rd and 4th order phase, hole position, width and depth are inputs to this setup.

FROG reconstructs the characteristics of a femtosecond laser pulse and we use seven DAZZLER figures of merit to evaluate the quality of the laser pulse. These are listed below with the six inputs from DAZZLER:

- DAZZLER settings (inputs): $\phi_2, \phi_3, \phi_4, r_H, w_H$, and d_H .
- FROG (figures of merit): $TBP_{FWHM}, TBP_{RMS}, e_\phi, s, k$.

Here ϕ_2, ϕ_3 , and ϕ_4 represent the second-order, third-order, and fourth-order phases of the laser respectively. In addition, r_H represents the central wavelength of the hole in the laser spectrum. The parameter w_H refers to the full width at half maximum (FWHM) of the hole. d_H represents the relative amplitude of the hole. The output parameters, T_{FWHM} and T_{RMS} represent the FWHM and root mean square (RMS) of the pulse duration, respectively. Similarly, the TBP_{FWHM} and TBP_{RMS} represent the FWHM and RMS of time-bandwidth product, respectively. The parameter e_ϕ represents the deviation of the laser spectral phase from a flat phase case, s and k are the skewness and kurtosis excess [40] of a random variable X , respectively. They are defined as

$$s(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}, \quad (1)$$

$$k(X) = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4} - 3, \quad (2)$$

where $\mathbb{E}[\cdot]$ represents the expectation operator, μ is a mean value of X , and σ is its standard deviation.

In general, any type of a laser pulse is described by the amplitude and phase. In the DAZZLER, the amplitude is determined by the hole. The hole is an area in the laser spectrum where we do not want the pulse to be amplified. The phase is determined by the phases at that particular hole. Such chirped pulse amplification (CPA) and hole detection has been reported by Strickland et al. [41]. The configuration of the hole is determined by three parameters: position, width, and depth. By adding a hole to the laser spectrum before sending it to the amplifier, one can produce a laser pulse with a large bandwidth, and thus, a short pulse duration.

The shape of the laser pulse is a Gaussian distribution with TBP_{FWHM} as the full width of half maximum of the distribution. TBP_{RMS} is the root mean square of the distribution. These parameters were used to calculate the pulse duration. Also, TBP_{FWHM} is the FWHM time-bandwidth product, and TBP_{RMS} is the RMS time-bandwidth product. Here, the time-bandwidth product is a parameter that characterizes the pulse shape, similar to the pulse duration. It is the product of the temporal pulse duration and the spectral bandwidth.

In this work, we implemented the NNs to control the ultrafast pulsed laser system and therefore the electron beam characteristics of a laser-driven accelerator.

3.3. Modeling the laser

The computations required for modeling in the FFNN include the backpropagation algorithm [42]. It is a gradient descent method that iteratively updates the weights of the NN by moving them in the

direction opposite to the gradient of a cost function until a minimum is reached. In general, the log-likelihood of the training outputs is conditional on the inputs.

In our particular application, it is commonly assumed that the training error has a Gaussian distribution [43]. Then, the output likelihood of a sample can be written as

$$p(y|\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{1}{2\sigma^2} \|y - z\|^2\right) \quad (3)$$

where $y \in \mathbb{R}$ is the desired output, z is the obtained output of the NN, which, since it is defined in the whole real axis, is implemented with a linear activation at the output of the NN.

Therefore, the log-likelihood of the full data set can be written as

$$\begin{aligned} J_{ML}(\theta) &= -\mathbb{E}_{\mathbf{x},y} \log p(y|\mathbf{x}) = \mathbb{E}_{\mathbf{x},y} \left(\frac{1}{2\sigma^2} \|y - z\|^2 + \frac{1}{2} 2\pi\sigma^2 \right) \\ &= \mathbb{E}_{\mathbf{x},y} (\|y - z\|^2) + \text{constant} \end{aligned} \quad (4)$$

which is simply the mean squared error (MSE).

We used the dataset of 10 000 points to train, validate, and test the NNs. The NNs have six inputs and one output. The data was divided into three parts: 70% of the data was used for training, 15% for validation, and 15% for testing.

Before the NNs can be developed, we need to plot the relationships between inputs and outputs to see if there are trends between the variables and detect possible multicollinearity. In our case, the correlations were not observed in almost all combinations of inputs and outputs, indicating that the system is very complicated. The best trend was observed for the laser pulse FWHM (output) versus the second-order phase (input), as shown in Fig. 4(a). On the other hand, Fig. 4(b) shows that there is no correlation between the FWHM for the third order.

After exploring the new statistical data, we developed two NN models that could establish the complex relationships between the input and output parameters of the laser to optimize the TBP_{RMS} . The script for retrieving and organizing the data and training the NNs was further improved. The script is used to retrieve, organize, and write the data to a spreadsheet in the correct format. This properly organized data can be used to develop input/output models. The models were developed using NNs with second, third, and fourth order input phases and hole position, width, and depth as input parameters. The output parameter is TBP .

Fig. 4(b) shows that there was no correlation for the third order. After exploring the new statistical data, we developed two NN models, that could establish the relationship between the input and output parameters. A script for retrieving and organizing the data and training the NNs was further optimized. The script is used to recover, organize, and write the data in a spreadsheet in the proper format. These properly organized data can easily be used to develop the input/output models. These models were developed using the NNs with second-, third-, and fourth-order input phases, and hole position, width, and depth as input parameters. The output parameter was TBP_{RMS} . This exercise aimed to improve the present and future performance of the laser system using ML techniques based on the instrumentation data.

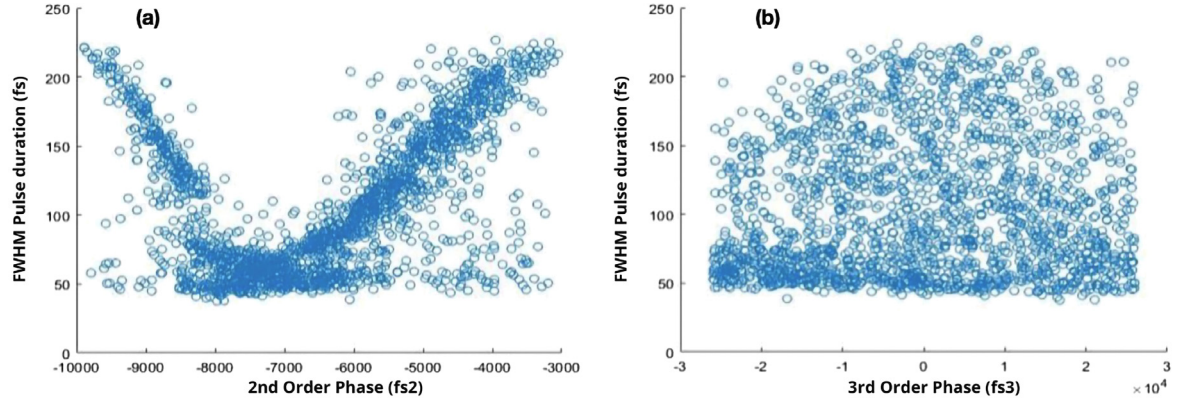


Fig. 4. (a) Trends between the second-order phase (input) and the FWHM pulse duration (output). (b) Trends between the third-order phase (input) and the FWHM pulse duration (output).

3.4. Training dataset

The input data includes 2nd, 3rd, and 4th order phases, hole position, hole width, and hole depth. Even though we neglect delay, which is the 1st order phase, the software treats 1st order phase as a delay. Therefore, the software was designed to always adjust the 1st order to be in the middle of the delay range to protect the device. Moreover, the range of the second order phase corresponds to a time delay between -10000 fs^2 and -3000 fs^2 . The third order phase corresponds, to a time delay between $-2.6 \times 10^4 \text{ fs}^3$ to $2.6 \times 10^4 \text{ fs}^3$. The fourth order phase corresponds, to a range between $-3.0 \times 10^5 \text{ fs}^4$ and $3.0 \times 10^5 \text{ fs}^4$. The range of the hole position was 750–825 nm, the range of the hole depth was 0%–100%, and the range of the hole width was 0–100 nm. Note that for the hole, these limits are not mandatory; however, it is not feasible if we have a larger width or position. For example, it would be redundant if the hole width was larger than the spectral bandwidth. The hole depth could only range from 0% to 100%.

3.5. Training and testing process

As mentioned above, the networks were trained using the Levenberg-Marquardt optimization algorithm [44,45]. The weights and biases were initialized using the Nguyen-Widrow algorithm [46]. It chooses values to randomly but uniformly distribute the active region of each neuron in the layer. Repetitive rows were removed to reduce redundancy.

The dataset was divided into three parts: 70% of the data was used for training, 15% for validation, and 15% for testing. We used the sequential order incremental adoption and the gradient descent with momentum weight and bias learning function.

To measure the performance of the trained networks, we used the MSE of TBP_{RMS} as the loss function:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\text{TBP}_{\text{RMS},i} - \widehat{\text{TBP}}_{\text{RMS},i})^2, \quad (5)$$

where N is the number of training samples, and $\widehat{\text{TBP}}_{\text{RMS},i}$ are the predicted values.

The computations were performed using MATLAB on the MacOS BigSur operating system with an Intel Core i9 processor clocked at 2.3 GHz. The training time was 6.67 s for NN1 and about 14 s for NN2.

The MSE on the test set, the slope of the regression curve, and the Pearson's R coefficients for both NNs are summarized in Table 1. The loss curves of NN2 for training, validation and test sets are shown in Fig. 5.

Fig. 5(a) shows the results of regression for NN2, which demonstrate that the computed values are highly correlated with the target values. Although there are several outliers, an overall trend can be clearly

Table 1

Regression results for two considered neural networks.

	NN1	NN2
MSE	0.0023	0.0014
slope	0.87	0.89
R	0.89913	0.94297

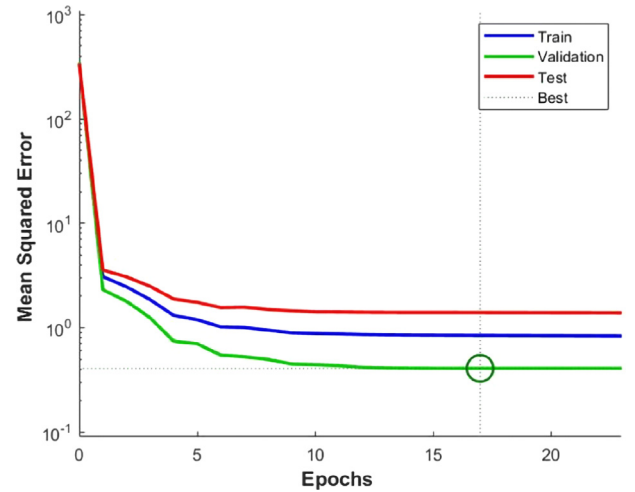


Fig. 5. Loss curves for training, validation, and test sets for the NN2. The best validation loss (MSE = 0.40755) was found at epoch 17.

Table 2

Comparison of the MSE (on the test set) and the wall clock inference time for two proposed neural networks versus the alternative regression methods.

	NN1	NN2	SVM	GA	RF
MSE	0.0023	0.0014	0.0059	0.0142	0.0025
Time (s)	0.10830	1.8107	2.7876	2.6586	0.2545

observed. The dashed line $Y = T$ is the desired output line passing through the origin. The linear regression fit is shown in the red line, its offset is 5.4 and the slope is 0.89.

The error histograms in Fig. 6(b) indicate that model has reached a reasonable accuracy. A few outliers might be associated with a lack of repeatability in the experimental results.

Table 2 presents a comparison of the MSE and the wall clock inference time of two neural networks (NN1 and NN2), support vector machine (SVM), genetic algorithm (GA), and random forest (RF) for regression. From the results, we can see that NN2 has the lowest MSE

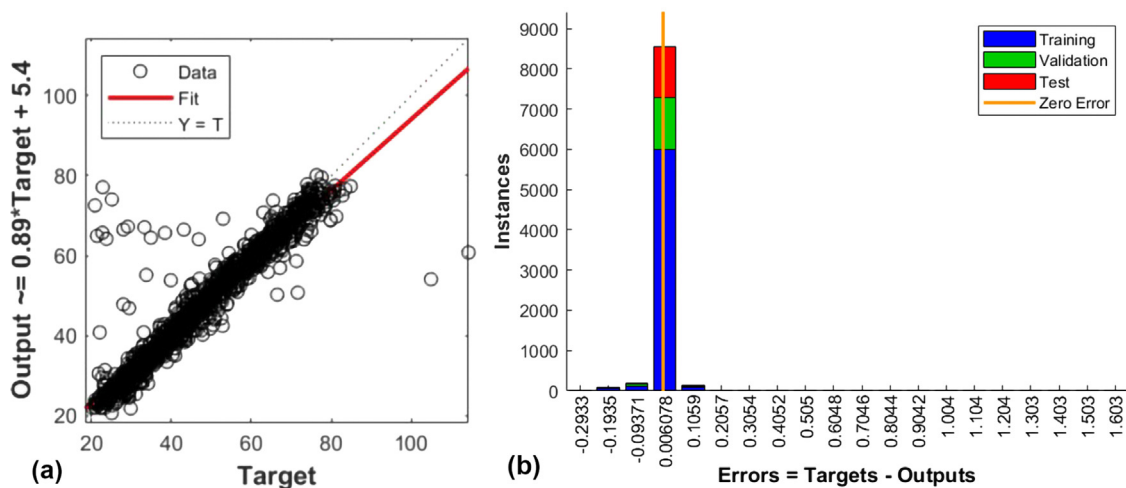


Fig. 6. Regression curves (a) and error histogram (b) obtained with the second NN after training on the dataset with 10 000 points.

among all the methods, with a value of 0.0014. This suggests that NN2 has the best performance in terms of accuracy. On the other hand, NN1 has a slightly higher MSE of 0.0023, but it has the fastest inference time of 0.10830 s. The SVM method has a relatively high MSE of 0.0059 and a moderate inference time of 2.7876 s. The GA method has a relatively high MSE and a relatively long inference time. The RF method has a relatively low MSE of 0.0025 and a fast inference time of 0.2545 s.

In conclusion, the results suggest that NN2 is the best performing method in terms of accuracy, while NN1 has a trade-off between accuracy and inference time. The RF method has a good balance between accuracy and speed, while the SVM and GA methods are relatively slower and less accurate. The choice of method depends on the specific requirements and limitations of the problem at hand, such as the trade-off between accuracy and computational resources.

4. Results and discussion

In this paper, we report two neural network models that establish the relationship between the input spectral amplitude and phase of the laser pulse and the output laser pulse shape. The uniqueness of our method is the ability to address the difficulties such as short time scales and nonlinearity in the system. We reported the optimization of the intensity of a femtosecond laser, which models the input-output relationship of the data. Using this principle, the input and output parameters of the system were interpolated to produce the required femtosecond laser power. Improvements in laser control through this optimization can directly lead to better electron beam control. The idea is that the improvements in the laser parameter control mechanisms (in this investigation, using the NNs) can lead to faster convergence to the optimal laser beam parameters that ultimately produce the desired particle beam parameters. In future work, we hope to define the desired particle beam parameters in the feedback loop and modulate the laser characteristics that produce the particle beams to achieve optimized particle beams.

With the help of NNs, the computationally expensive optimization task is performed during the training. Once the model is trained, it can be used as a universal parameter optimizer in many similar situations. Nevertheless, the acquisition of the training dataset was the most challenging task. Although the specific amount of data depends on many factors, tens of thousands of points are often regarded as the minimum. Therefore, collecting suitable training data can be a lengthy and challenging process, which, in the case of laser machining, as one example, would require the manufacturing of tens of thousands of machined structures of various shapes, sizes, and even materials [47]. Several progressive methodologies can be adopted to mitigate the problems [48] that can occur during the system process.

Developments in the field of NNs have made it feasible to solve complex structures and train our algorithms and applications to cope with these challenges [49]. Advancements in computational resources have also significantly increased the speed of training, allowing much larger datasets to be used in practice. Furthermore, with huge computational storage, it is simple to store and gather large datasets for training.

Beneficial co-developments in related fields, such as stochastic optimization and machine learning, have helped us augment advanced algorithms with nonlinear and optimal control for successful deployment of NNs. High-performance computing (HPC) infrastructure is now available in national laboratory systems, such as up to exascale at the Argonne Leadership Computing Facility [50], as well as computing resources at universities and in industry, facilitating the use of ML. The greater availability of large datasets and a host of new open source and software libraries have also helped to improve research on existing ML techniques and facilitate rapid prototyping of new algorithms.

5. Conclusions and future scope

In summary, we have modeled the relationship between the experimental DAZZLER settings (inputs) and the temporal output width of the femtosecond laser pulse using machine learning techniques. The parameters used for the model were the second, third, and fourth-order input phases (ϕ_2, ϕ_3, ϕ_4); hole position (r_H), width (w_H), and depth (d_H) as input parameters, and TBP_{RMS} as the output. The parameters represent the device characteristics; hence, the feature selection is good and can be extended to any other laser system of a similar nature. The input of the DAZZLER under open-loop conditions and the corresponding output of the FROG was used as the input and output, respectively, to train the NN.

After improvements in data selection and tuning, the NNs were used for modeling to improve the temporal width of the femtosecond laser pulse and other output parameters. Remarkable results were achieved. ML techniques can be used to meet the requirements of laser operation through better control of the electron beam. These and similar techniques are also applicable to many other complex systems, including research infrastructures (RIs) and energy infrastructures for the public good [51], research facilities [52,53], the design, control and application of quantum information systems [54], and better low level radio frequency control [55].

The future scope of this work includes investigating the dependence on the number of layers and neurons, different training and testing algorithms, different types of NNs, and many other parameters to evaluate the performance of our models and their variants. We will automate the process of optimizing the number of layers, number of nodes, and choice of learning rate. We plan to test this approach experimentally

at the University of Michigan in 2023 and obtain additional datasets to refine our model to reliably model the complex system. Because large amounts of labeled experimental data are typically required, data collection and labeling in the case of NN is a challenge. In addition, in the future, we may use the cascade feedforward network, which has additional connections from the input to each layer and from each layer to all subsequent layers. Finally, we are using and expanding these methods in the development of a drive-laser-based electron-beam emittance ML control of the Accelerator Test Facility's (ATF's) main linac [56] as well as the tuning and control of a MeV Ultrafast Electron Diffraction experiment at the ATF [57,58].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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